

Synergistic Promotion of Single-Atom Co Surrounding a PtCo Alloy Based On a g-C₃N₄ Nanosheet for Overall Water Splitting

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Cite This: *ACS Catal.* 2022, 12, 6958–6967



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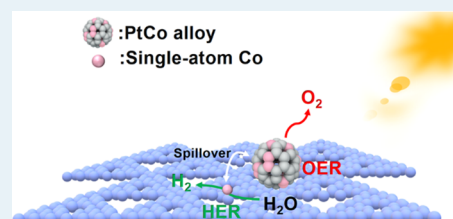
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ABSTRACT: Atomically dispersed catalysts with maximum efficiencies for atom utilization have emerged as a frontier in the catalytic field. However, single-component monatomic catalysts are rarely reported for the overall photocatalytic water splitting reaction (OWS). Herein, a two-component synergistic photocatalyst was designed that contained a dispersion of single-atom Co (Co_{SAs}) centers and PtCo alloy nanoparticles (Nps), which were supported over C₃N₄ nanosheets. Among these components, Co_{SAs} centers act as highly active sites for the hydrogen evolution reaction (HER), and PtCo alloys act as highly active sites for the oxygen evolution reaction (OER). A synergistic effect between them occurs when the two different reactive centers are combined, indicating that there may be a spillover phenomenon of protons or hydroxyl groups between the Co_{SAs} centers and PtCo alloy NPs. The synergistic promotion of Co_{SAs} centers and PtCo alloys bridges the gap between achieving maximum atom utilization and optimal bifunctional activities for OWS reactions. This combination provides a promising model for the development of OWS by atomically dispersed catalysts.

KEYWORDS: single atoms, overall water splitting, synergistic effects, spillover, photocatalysis



1. INTRODUCTION

Sunlight-driven water splitting that produces hydrogen and oxygen gas evolution on a large scale is an ideal strategy that may satisfy the need for clean energy and solve the fossil fuel crisis.^{1,2} However, overall water splitting, without the elimination of sacrificial reagents or the need to apply an external electrical bias, requires synergistic active sites that connect the compartmentalized and spatially separated hydrogen and oxygen evolution reactions. By itself, this generates more critical requirements for the synthetic engineering of photocatalysts.^{3,4} Many photocatalysts possess the intrinsic disadvantages of wide band gap energies, easy deactivation of reaction sites, and low-efficiency electron–hole separation processes in the overall progression of water splitting.⁵ Therefore, the development of a photocatalyst for overall water splitting has always been one of the most challenging tasks in the field of catalyst development and is viewed as the holy grail in the field of modern chemistry.^{6,7}

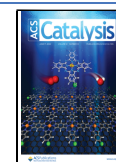
Recently, atomically dispersed catalysts with the maximum efficiencies for atom utilization have emerged as a new frontier in the catalytic field. Some have proved to have excellent catalytic activities and selectivities for the hydrogen evolution reaction (HER), the oxygen evolution reaction (OER), the nitrogen reduction reaction (NRR), or other important reactions.^{8–10} For example, single atoms (SAs) coordinated by heteroatoms on N_xC_y substrates (N_xC_y = g-C₃N₄, amorphous carbon, graphene, and carbon nanotubes) have been widely explored as efficient catalysts for many photo-

chemical and electrochemical reactions.¹¹ Among them, g-C₃N₄ not only has a visible-light response, a high chemical stability, and appropriate C_{2p} and N_{2p} orbital bands that straddle the water splitting potentials but it also possesses more N coordination (over a large pore volume) that prevents metal agglomeration through improving the metal–support interaction, which has been widely used as a photocatalyst for water splitting.^{12–15} The metal–support interaction plays a key role in regulating the charge density and d orbital components of the loaded metal species, thereby limiting the size of metal species to the atomic level. However, only one kind of single atomic site cocatalyst is rarely reported to be successful in the overall reaction of photocatalytic water splitting because (1) the thermal stability of the exposed single atoms is poor and they easily migrate and agglomerate during the photocatalytic process, especially during the oxidation process,¹⁶ and (2) SAs catalysts are limited by a size effect and show a strong selectivity for forming a single product. Thus, it is difficult to find SAs catalysts that act as bifunctional active sites for hydrogen and oxygen evolution simultaneously.^{7,17}

Received: February 11, 2022

Revised: May 14, 2022

Published: May 30, 2022



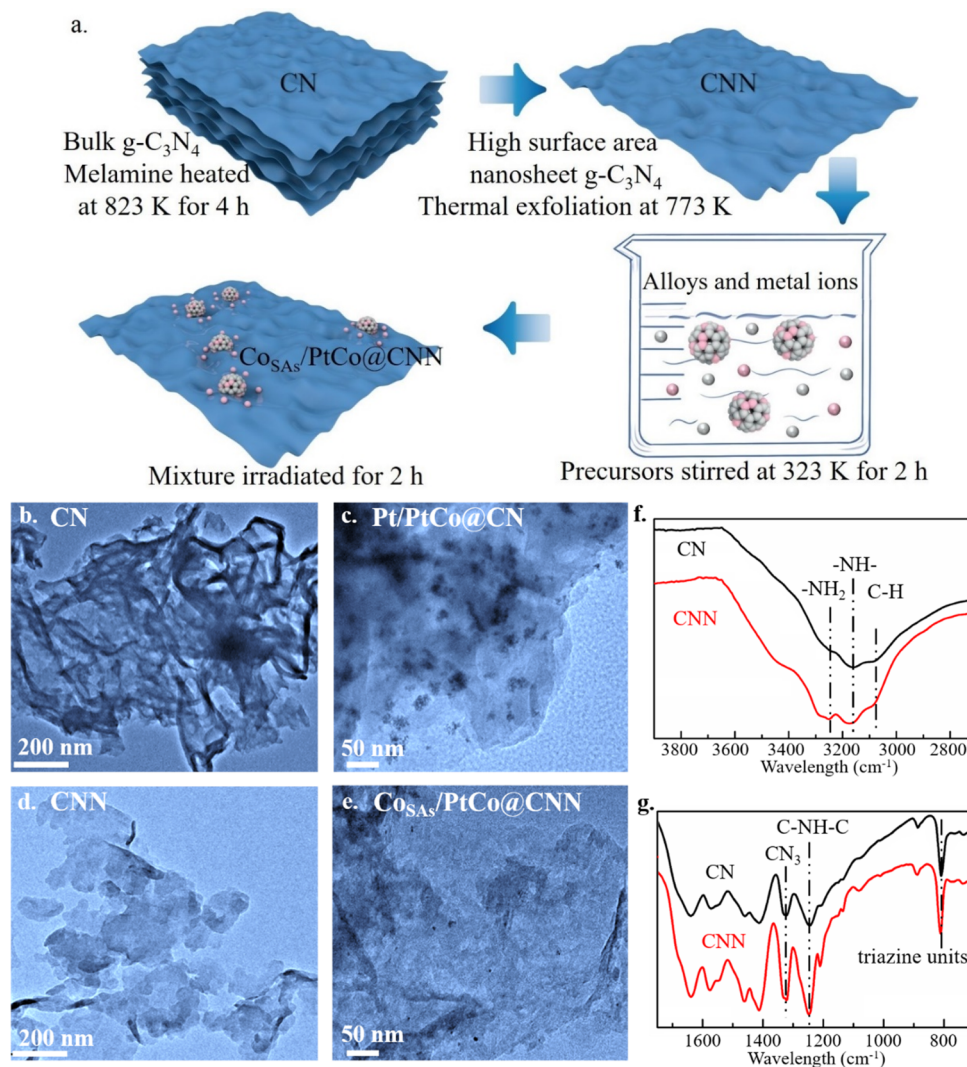


Figure 1. (a) Schematic illustration of the synthetic strategy of producing the $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ sample. TEM images of (b and c) the CN substrate (representing the bulk sheet forms of CN and $\text{Pt}/\text{PtCo}@ \text{CN}$) and (d and e) the CNN substrate (representing the nanosheet forms of CNN and $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ with high surface areas). Fourier transform infrared (FTIR) spectroscopy results of CN and CNN substrate samples for scan ranges between (f) 2800–3800 and (g) 800–1600 cm^{-1} .

An important fact borne out in the literature is that conventional $\text{g-C}_3\text{N}_4$ is severely limited by sluggish overall kinetics in photocatalytic water splitting due to the low efficiency of the electron–hole separation.¹⁸ This is because of the lack of surface redox active sites (the oxidation of water to oxygen requires the transfer of four electrons) causes the electron–hole separation to be much more energetically demanding than hydrogen production.¹⁹ Therefore, multifaceted strategies have recently been developed to activate polymeric carbon nitride semiconductors for overall water splitting to produce H_2 and O_2 without any sacrificial agent; these include doping modifications, composite or molecular designs, and band structure engineering.^{18–20} Unfortunately, the multistep construction of various heterostructures or metal doping centers will change the original framework of C_3N_4 , making the process of catalyst synthesis more cumbersome and expensive. The resulting complex interface environment is not conducive to the study of the structure–activity relationship.⁶ Thus, it remains challenging to apply SAs to achieve overall photocatalytic water splitting without changing the intrinsic structure of $\text{g-C}_3\text{N}_4$.

Focusing on the application of single atoms to the overall water splitting reaction, we conceived of a two-component synergistic photocatalyst that contained both Co SAs and a PtCo alloy, which were supported over nanosheet C_3N_4 . We denote this design as the $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ catalytic system. The $\text{g-C}_3\text{N}_4$ photocatalyst used in this work is nanosheet C_3N_4 with a surface area of 277.86 m^2/g . After the preliminary nucleation and photoreduction of Pt and Co metal promoters, most of the Co SAs actively surround PtCo alloy nanoparticles. Combining electrochemical data from these reference catalysts and theoretical calculations, we determined that the Co SAs act as the active sites for the HER, and the PtCo alloys act as the active sites for the OER. We discovered that this two-component synergistic photocatalyst exhibited the SAs advantages of maximum atom utilization and efficiency. This study highlights the structure–activity relationship in overall water and water–half splitting reactions. It also shows how to greatly improve the degradation performance of organic compounds (enhanced 24 $\times$) and provides a promising model for the development of single-atom catalysts.

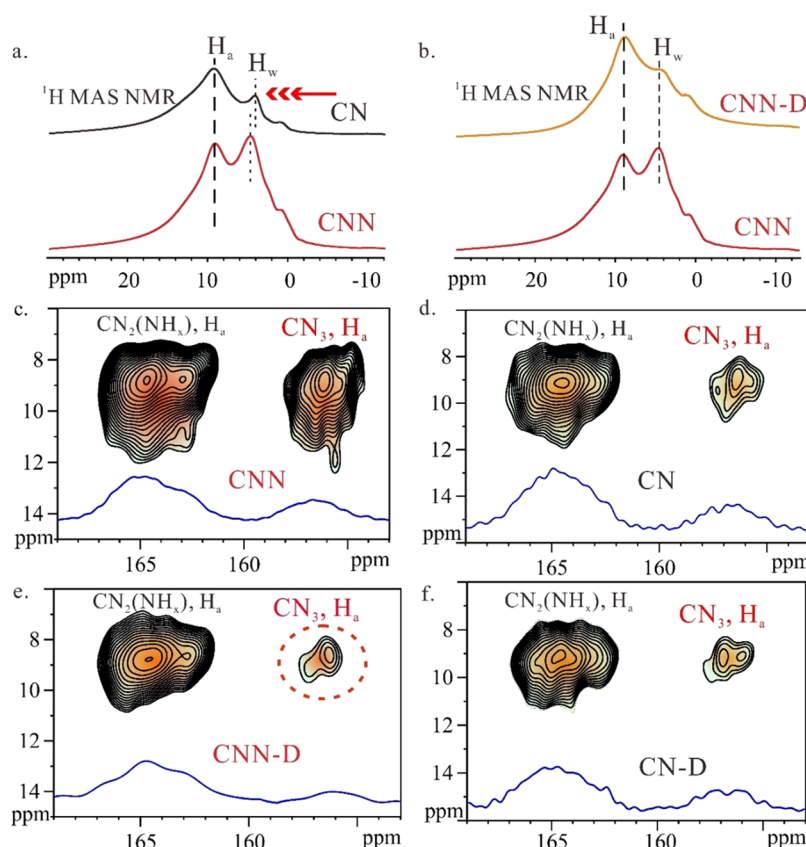


Figure 2. Solid-state nuclear magnetic resonance (SSNMR) spectra of the as-synthesized bulk graphitic carbon nitride (CN) and high-surface-area nanosheet $g\text{-C}_3\text{N}_4$ (CNN) samples obtained under equivalent conditions before and after D_2O exchange. The D_2O -treated CN and CNN samples are denoted as CN-D and CNN-D, respectively. (a and b) One-dimensional ^1H magic angle spinning (MAS) NMR spectra. Panel a shows ^1H MAS NMR spectra for CN and CNN samples. Panel b shows ^1H MAS NMR spectra for deuterium-exchanged samples of CNN-D and CN. Two-dimensional (2D) ^1H – ^{13}C heteronuclear chemical shift correlation (HETCOR) NMR spectra of (c) CNN, (d) CN, (e) CNN-D, and (f) CN-D samples.

2. RESULTS AND DISCUSSION

To develop nanosheet $g\text{-C}_3\text{N}_4$ with a high surface area and high pore volume (denoted as CNN) for improved overall water photocatalysis, nanosheet $g\text{-C}_3\text{N}_4$ (CNN) was synthesized by a facile and scalable two-step calcination procedure (see Figure 1a and section 4.1 for details). Utilizing a large-diameter crucible with bare leakage, the method produced a product yield of $\sim 8\%$. The microstructures of the CN (original $g\text{-C}_3\text{N}_4$) and CNN nanosheet samples were initially investigated by TEM.

Figure 1b shows that the CN sample is composed of large wavy layers in a bulging and continuous structure. The enlarged view in Figure 1c suggests that metal particles aggregate on either the surface or the subsurface of bulk CN after metal loading (Pt/PtCo@CN sample as the reference catalyst, see section 4.1 for details). After the two-step calcination, the resultant CNN sample is transformed to thin, loose, and soft silk-like nanosheet structures (Figure 1d). Our data suggest that the calcination method mediates the curling of CN layers, which results from a decrease in the surface tension. The outcome seems to leave the metal particles more evenly and stably loaded on the surface (enlarged view in Figure 1e, the $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ sample) in a manner that requires the formation of exfoliated carbon scrolls during graphite peeling. Moreover, this morphology change prominently increases both the surface area and the pore volume from $44.78 \text{ m}^2/\text{g}$ and $0.27 \text{ cm}^3/\text{g}$ to $277.86 \text{ m}^2/\text{g}$ and 1.65

cm^3/g , respectively. This transformation is helpful to limit the particle size of the loaded metal, confining the size to the single-atom level. As shown in Figure 1f and g, the Fourier transform infrared (FTIR) spectra of the CN and CNN samples were separated into two ranges. One scan shows the range between $2800\text{--}3800 \text{ cm}^{-1}$, and the second scan shows the ranges between $800\text{--}1600 \text{ cm}^{-1}$. We note that the overall FTIR spectra of the CN and CNN samples are fundamentally similar, which is indicative of a common structural composition.²¹ The absorption peaks observed at $\sim 809 \text{ cm}^{-1}$ for the CN and CNN samples were attributed to the intense stretching vibrations of the C–N bonds between heptazine units. Absorption peaks associated with the typical stretching vibrations of the aromatic rings of C–N heterocycles were observed in the wavenumber range of $1200\text{--}1480 \text{ cm}^{-1}$. The stretching vibrations of C–N–(–C)–C bonds were observed at 1323 cm^{-1} , and those of the C–NH–C bonds were observed at 1243 cm^{-1} . For photocatalytic reactions that are based on $g\text{-C}_3\text{N}_4$ catalysts and derivatives, the –NH defect is generally present at hydrophilic active sites that facilitate photocatalytic water splitting.^{22–24} As shown in Figure 1g, the vibrations in the C–NH–C bonds of the CNN sample are significantly stronger than those in the CN sample, indicating that CNN has a high concentration of –NH– defect sites that might enhance the photocatalytic activity of the water molecules. Here, the increase in the surface area and the

pore volume introduces more beneficial defects in the nanosheet CNN sample.

Solid-state nuclear magnetic resonance (SSNMR) spectroscopy provides detailed information on the relationship between the H-containing groups of $g\text{-C}_3\text{N}_4$ photocatalysts and water splitting sites at the atomic level.^{25,26} Figure 2a presents the one-dimensional (1D) ^1H magic angle solid (MAS) NMR spectra of the CN and CNN samples. ^1H NMR spectra of the CN samples show two strong peaks at chemical shifts of 3.91 (H_w) and 8.95 ppm (H_a); these signals arise from residual water and amino groups, respectively. ^1H NMR spectra of the CNN sample show residual water and amino group peaks at chemical shifts of 4.42 (H_w) and 8.92 ppm (H_a), respectively. The most obvious difference between the ^1H NMR spectra obtained for the CN and CNN samples is the chemical shifts and relative strengths of the H_w signal. For the CNN samples, the chemical shift of H_w moved 0.51 ppm downfield compared to the H_w peak for the CN sample. At the same time, the intensity of the H_w peak for CNN is greater than that for CN. The observed shift and increase in the peak H_w intensity of the CNN sample suggest that it has stronger hydrophilicity and a newly formed hydrogen bond network. These changes are presumably due to the exposure of more NH_x defects and benefit small-molecule adsorption (see Figure 1f and g). Figure 2b presents the 1D ^1H MAS NMR spectra of the CNN sample obtained before and after a D_2O treatment (denoted as CNN-D). When the residual water in the CNN sample is deuterated via the D_2O treatment, the H_w signal of CNN-D weakens significantly (as shown in Figure 2b). This indicates that CNN samples possess the dual advantages of facile adsorption and desorption for water molecules. In contrast, the lack of a significant change in the H_w signal intensity or its position from ordinary CN samples after the D_2O treatment indicate that was no obvious structural change due to hydrogen exchange (Figure S1).

An important consideration for elucidating the amino structure of the photocatalytic active sites of $g\text{-C}_3\text{N}_4$ is the capability for extensive proton exchange with the ^2H atom of D_2O after the deuterium exchange. The degree of exchange can be effectively used to evaluate the relative activities of hydrophilic sites in the photocatalytic splitting of water. Figure 2 c–f presents the two-dimensional (2D) ^1H – ^{13}C heteronuclear chemical shift correlation (HETCOR) NMR spectra. The correlation peaks originate from coupled proton signals that arise from the interaction of single carbon atoms with a proton or a plurality of protons in close proximity. Panels c and d in Figure 2 show two correlation peaks from the CNN and CN spectra, respectively. One is the correlation peak between the ^1H signal at ~ 9.0 ppm and the ^{13}C signal at ~ 164 ppm (denoted ($\text{CN}_2(\text{NH}_\text{x}, \text{H}_\text{a})$), and the other is the correlation peak between the ^1H signal at ~ 9.0 ppm and the ^{13}C signal at ~ 156 ppm (denoted ($\text{CN}_3, \text{H}_\text{a}$)). The difference is that the correlation signal ($\text{CN}_3, \text{H}_\text{a}$) for the CNN sample is obviously stronger than that for the CN sample, which indicates that the bulk structure of CNN sample has more amino defects (NH_x). This is consistent with the observation that a high surface area and an increase in the pore volume (as seen in Figure 1f) produce more beneficial defects that activate the nanosheet CNN samples. After deuterium exchange (see Figure 2e), the correlation of the $\text{CN}_3, \text{H}_\text{a}$ peaks of the CNN-D samples decreased significantly, indicating that there was strong proton exchange between the marginal amino group (H_a) and deuterated water. In contrast, the evidence of proton exchange

in the CN-D samples between H_a and deuterated water exhibited almost no change before and after the deuterium treatment (see Figure 2f). By selective deuteration via proton exchange from D_2O to the active sites of photocatalysts, we found that the CNN samples exhibited greater accessibility and a better affinity for water as well as a stronger propensity to develop amino defects compared to the original CN catalyst.

Having characterized the CNN nanosheet and shown its greater surface area and pore volume compared to the CN multisheet samples, we proceeded to load the metal centers onto the CNN surface after several crucial treatments, such as preliminary nucleation and photoreduction deposition (denoted as $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ catalysts, see Figure 1a and section 4.1 for details). Previous studies demonstrated the cationic nature of metal atoms isolated on $g\text{-C}_3\text{N}_4$ and its derivatives.²⁷ Similarly, XPS analyses revealed two valence states of Co in the $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ sample; these were tentatively assigned to Co^0 (777.8 eV) and Co^{2+} (781.2 eV) complexes based on formal charges (Figure S2a). The synthesis process required the anaerobic isolation of the metal precursors during the thermal reduction and photoreduction. We speculate that the Co^{2+} component mainly coordinates with N atoms in the skeleton pores to form a CoN_x structure in agreement with its known electronegativity, while the Co^0 species mainly exist in the form of alloys or Co particles. We found it noteworthy that Pt species were mainly concentrated in the Pt^0 state. The Pt^0 species that was loaded on the CNN surface could be integrated on its supports in the form of alloys or Pt particles, as shown by the XPS analysis (see Figure S2b). The XRD analysis confirmed the formation of the $\text{Pt}_\text{x}\text{Co}$ alloy structures in the $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ catalyst. As shown in Figure S2c, the diffraction peaks at $2\theta \sim 46^\circ$ are consistent with those of $\text{Pt}_\text{x}\text{Co}$ alloys. These data indicate that Pt^0 and Co^0 species are predominant in the alloy structure.^{28,29}

We also synthesized the reference $\text{Pt}/\text{PtCo}@ \text{CN}$ catalyst (original $g\text{-C}_3\text{N}_4$) and the reference $\text{Pt}/\text{PtCo}@ \text{CNN}$ catalyst (thermal polymerization method). These two reference catalysts cannot produce monatomic active sites in thermodynamics. (see section 4.1 for details)^{5,17} Compared to the $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$, a new diffraction peak was observed at 39.9° with a Pt fcc structure (ICSD 64912) that corresponded to the (111) planes, indicating that Pt particle formation took place on the original $g\text{-C}_3\text{N}_4$ via the thermal reduction method (reference $\text{Pt}/\text{PtCo}@ \text{CN}$ and $\text{Pt}/\text{PtCo}@ \text{CNN}$ catalysts). In addition, XPS analysis of the reference $\text{Pt}/\text{PtCo}@ \text{CN}$ catalyst also demonstrated the existence of $\text{Pt}_\text{x}\text{O}_2$ species (see Figure S3). Figure S2d shows the steady-state photoluminescence (PL) emission spectra of $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ and its reference catalysts. Significant decreases in the PL occur on $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ and $\text{Pt}/\text{PtCo}@ \text{CNN}$, both of which are based on nanosheet CNN. That is, the intrinsic radiative recombination of photogenerated electron–hole pairs in $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ and $\text{Pt}/\text{PtCo}@ \text{CNN}$ is substantially inhibited by the modification of the substrate into a structure with a high surface area and a greater pore volume. This could only result from a stronger interaction between a loaded transition metal and the CNN host. The time-resolved PL spectra show that $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ has charge carriers with a longer average lifetime (τ_avg) than $\text{Pt}/\text{PtCo}@ \text{CN}$ (see Table S1). This change indicates that there is greater charge transfer in $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$, which was also demonstrated through photocurrent measurements (Figure S2e). $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ exhibits a

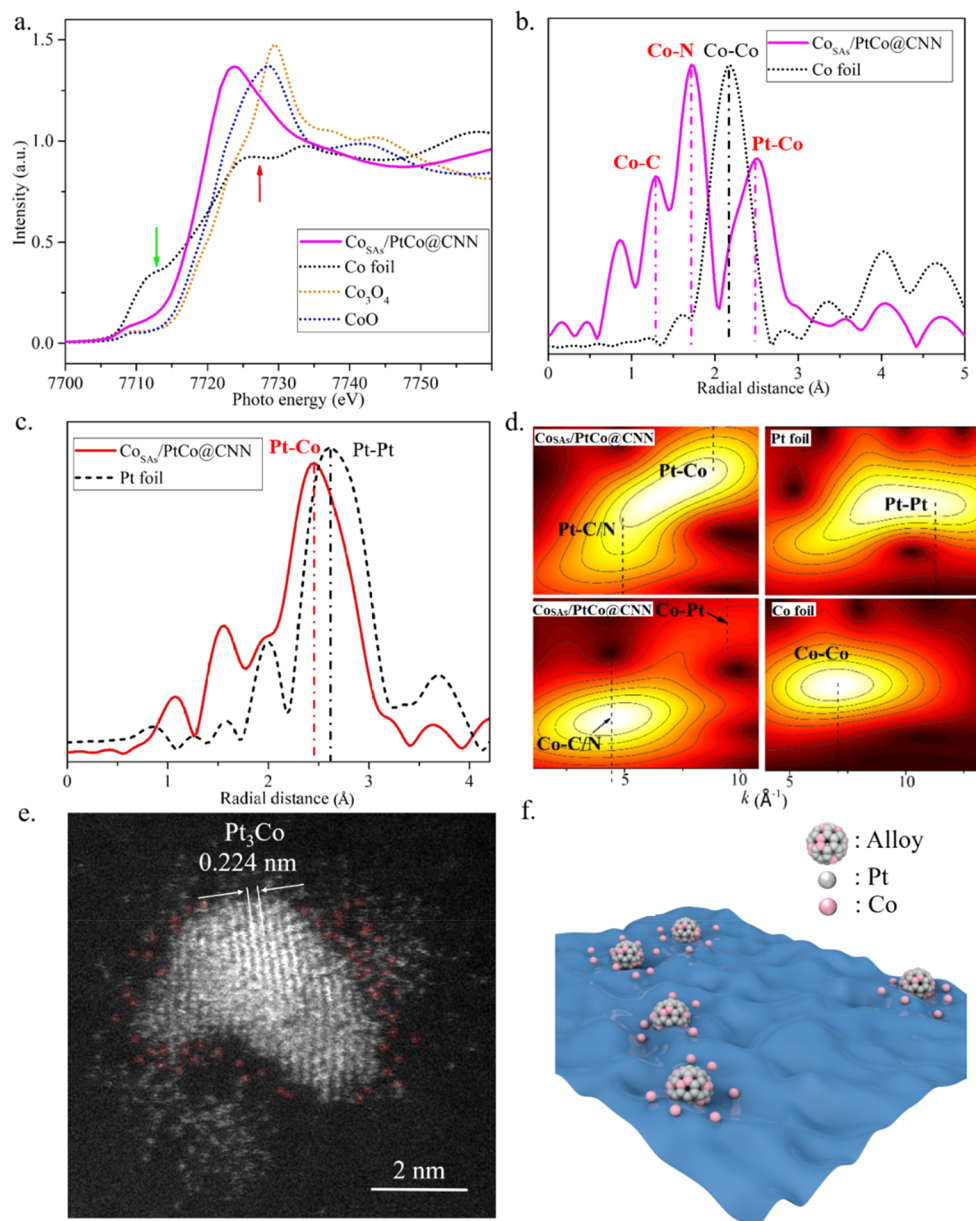


Figure 3. (a) XANES spectra taken at the Co K edge, with metallic oxide and metal foils as references. Radial distribution functions (RDFs) of the catalyst and reference samples from the EXAFS data taken at (b) the Co K edge and (c) the Pt L₃ edge. The RDFs were obtained after the Fourier transformation of the k^2 -weighted χ -function derived from the background-removed EXAFS data. (d) The wavelet transform (WT) contour plots of Pt L₃ edge absorbance spectra from CoSAs/PtCo@CNN and Pt foil samples and Co K edge spectra from CoSAs/PtCo@CNN and Co foil samples. (e) HAADF-STEM image of the CoSAs/PtCo@CNN catalyst with the measured lattice line spacing. Selected Co atoms (red circles) are identified. (f) Schematics of the CoSAs/PtCo@CNN catalyst showing the coexistence of PtCo nanoparticles and Co SAs species.

much greater photocurrent than the Pt/PtCo@CN catalyst based on original g-C₃N₄.

To gain further insight into the coordination chemistry of Pt and Co metals, we investigated the CoSAs/PtCo@CNN catalyst using X-ray absorption near edge structure (XANES) spectroscopy. As shown in Figure 3a, the XANES analysis demonstrated that the intensity of the Co peak at 7713 eV (green arrow) observed in the Co metal was reduced sharply in CoSAs/PtCo@CNN. By contrast, the peak at 7727 eV (red arrow in CoSAs/PtCo@CNN) was enhanced compared that of the Co foil, indicating the hybridization of Co 4s and 4p orbitals that occurred through combination with Pt electrons in the alloy in addition to the formation of Co(II)N_x coordination centers, as demonstrated by XPS.³⁰ The extended

X-ray absorption fine structure (EXAFS) analysis demonstrates the loss of Co–Co bonds, as evidenced by increased peaks of the Pt–Co bond (at 2.48 Å) and the Co–N bond (at 1.62 Å) in Figure 3b. This feature indicates the formation of the Pt_xCo alloy and N-ligated Co(II) structures (see Figure 3b). Alloy formation was further confirmed by a strong Pt–Co interaction peak at 11576 eV (blue arrow, seen in the XANES spectra of Figure S4). Similar alloy formation is also visible in the EXAFS spectrum of Figure 3c. The EXAFS spectrum at the Pt L₃ edge shows a 0.2 Å shift in the radial distribution of electrons compared to that of the bulk Pt foil, indicating the formation of Pt–Co bonds (2.48 Å), as confirmed by XRD and XPS results for the Pt species in this alloy. In addition, an intensity maximum at about 9 Å^{−1} can be

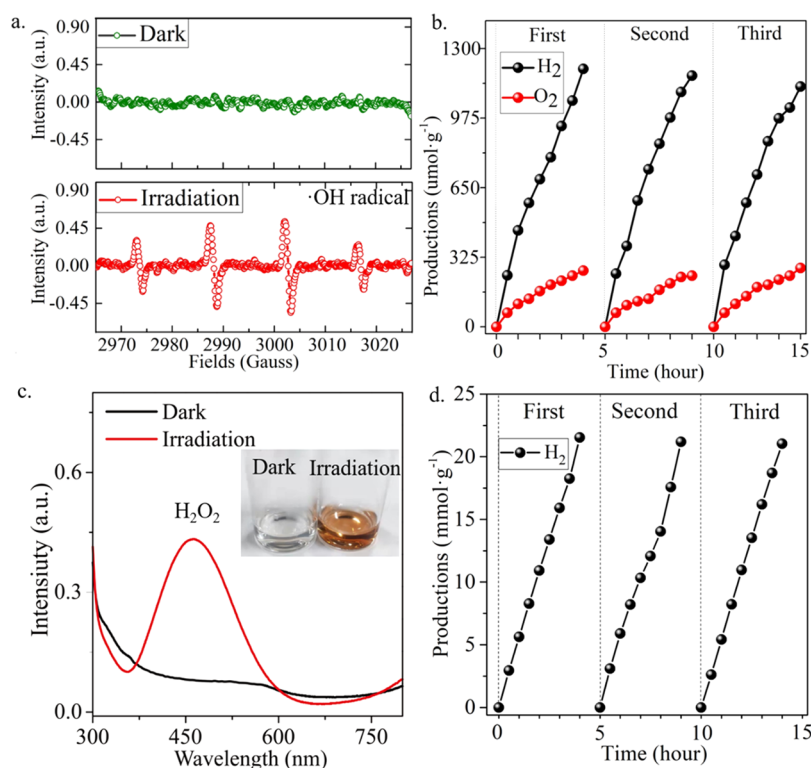


Figure 4. Photocatalytic measurements for overall water splitting. (a) In situ EPR spectra of the DMPO-OH spin adduct obtained in a $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ suspension. (b) Cycling of photocatalytic overall water splitting over the $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ catalyst. Each cycle started with a half hour of ultrasonic and vacuum pumping. Reaction conditions were as follows: 20 mg of the photocatalyst and 50 mL of pure water were illuminated under UV-visible light irradiation ($\lambda > 300$ nm). (c) The relative light absorption of H_2O_2 over the $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ suspension during photocatalytic overall water splitting. The production of H_2O_2 can be measured by UV-vis spectrophotometry. In an acidic environment (pH 2), H_2O_2 oxidizes Fe^{2+} (introduced from FeCl_2) to Fe^{3+} by , then Fe^{3+} combines with sufficient KSCN to form $[\text{Fe}(\text{SCN})]^{2+}$. $[\text{Fe}(\text{SCN})]^{2+}$ has a maximum absorption peak at 476 nm in the UV-vis spectrum. (d) Cycling of photocatalytic H_2 evolution over the $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ catalyst (20 mg), H_2O (40 mL), and TEOA (10 mL) under visible-light irradiation ($\lambda > 400$ nm).

seen from the wavelet transform (WT) contour plots of the Pt species from $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ in Figure 3d, which arises from Pt-Co coordination. Compared with the WT plots of the Pt foil, a weak Pt-C/N signal was detected at about $5 \text{ \AA}^{-1}$, and no Pt-Pt signal was detected. The EXAFS fitting results (Table S2) indicate that the Pt atoms are coordinated by Co atoms in the first shell, and some Pt atoms localized at the bottom layers of the alloy appear to interact with N in the CNN component. In the Co coordination in $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ (Figure 3d and Table S3), Co atoms are dispersed in monometallic sites, and the central Co atom is coordinated by four N atoms in the first shell. A small amount of the Co SAs may be coordinated by long-distance Co-N-C.

To elucidate the structural relationship of Pt_xCo alloys and $\text{Co}(\text{II})\text{N}_x$ sites, aberration-corrected high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) was carried out. Figure 3e isolates a single nanoparticle that has crystalline planes with 0.224 nm spacing (identified as the Pt_3Co plane) and is surrounded by numerous isolated metal atoms.³¹ Combined with XANES analysis, the nanoparticles (NPs) and isolated metal atoms were assigned to Pt_xCo alloys and single Co atoms, respectively. Figure 3f schematically illustrates the compositions of the $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ catalysts. First, most Pt atoms are involved in the formation of a PtCo alloy that is randomly distributed on the CNN surface. Cobalt atoms are mainly involved in two CNN-supported structural forms. In addition to their involvement in the formation of the bimetallic alloys, additional Co atoms are

unevenly scattered around the PtCo alloy clusters in the form of single atoms. We observed a small number of single Co atoms that were distant from NPs (see Figure S5). Collectively, all these forms constitute the $\text{Co}_{\text{SAs}}/\text{PtCo}$ structure.

The heretofore disclosed $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ catalyst was used to generate the product in the overall water splitting reaction under UV-visible light irradiation without the use of any electron sacrificial agents. The characteristic signals of $\cdot\text{OH}$ (hydroxyl radical) in the suspension were observed through in situ EPR spectroscopy (Figure 4a). This strong $\cdot\text{OH}$ signal suggests that the pathway involves the single electron oxidation of water to produce $\cdot\text{OH}$. Under the same experimental conditions, no $\cdot\text{OH}$ signal was detected from either the Pt/PtCo@CNN or Pt/PtCo@CN reference catalyst (Figure S6). To gain more insight into the overall water splitting, we further synthesized two single-component catalysts, namely $\text{Co}_{\text{SAs}}@CNN$ and PtCo@CNN , by loading single-atom Co and a PtCo alloy onto nanosheet CNN in separate samples (see Figures S7–S9 for characterization data and section 4.1 for details). The experimental results show that the PtCo nanoparticles of the PtCo@CNN and $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ catalysts are consistent in particle size (3.73 versus 3.81 nm), lattice parameters (0.224 versus 0.224 nm), and valence states of the Pt and Co elements (Pt^0 and Co^0 in PtCo@CNN). This proves that PtCo@CNN possesses a structure similar to that of the PtCo alloy in $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$. The $\cdot\text{OH}$ signal was still not detected over the $\text{Co}_{\text{SAs}}@CNN$ and

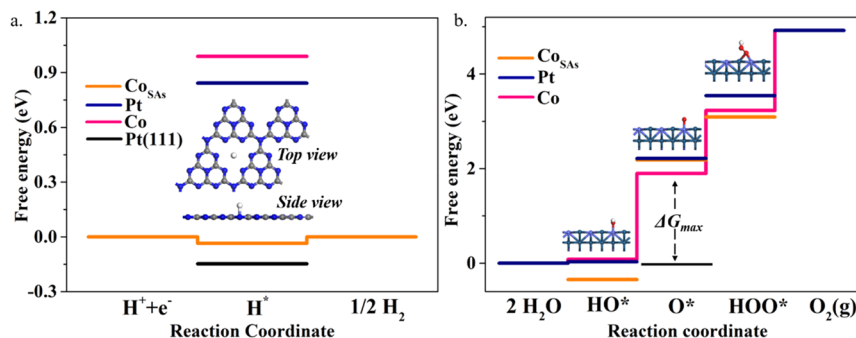


Figure 5. Free energy diagram for (a) the HER and (b) the OER under standard conditions for the $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ catalyst. The elementary reaction is characterized by the Gibbs free energy change ΔG_{max} which represents the rate-limiting step for photocatalytic overall water splitting.

PtCo@CNN catalysts (Figure S6). Therefore, we speculated that the generation of a large amount of $\bullet\text{OH}$ groups in the pure water environment was mainly derived from a synergistic effect of single-atom Co and PtCo alloys at the CNN interface over the $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ catalyst. This may indicate that there is a spillover phenomenon of protons or hydroxyl groups between the Co_{SAs} and PtCo alloys.^{32,33}

As shown in Figure 4b, $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ exhibits a high activity for H_2 gas production (up to $300.9 \mu\text{mol/h}\cdot\text{g}$) and an activity of $65.6 \mu\text{mol/h}\cdot\text{g}$ for O_2 production in the overall water splitting reaction. The reason that the stoichiometric ratio of H_2 and O_2 is not 2:1 in overall water splitting is that some oxygen atoms exist in the form of H_2O_2 in an aqueous solution. As shown in Figures 4c and S10, the H_2O_2 product was observed in the overall water splitting reaction. Within 4 h, the average ratio of H_2 , H_2O_2 , and O_2 products was 2:1.05:0.44. The H/O ratio of all photocatalytic products is 2:0.97, which was very close to the stoichiometric ratio of 2:1. After the catalyst was used for three cycles, the XRD diffraction peak was retained, and Co^0 on PtCo alloy maintained its stable structure. The Co^0 species still accounted for 14.3%, which is consistent with the proportion of the fresh $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ catalyst (see Figure S11). Neither H_2 nor O_2 was detected from the comparison Pt/PtCo@CNN and Pt/PtCo@CNN samples, which is consistent with the analysis of the EPR data. Moreover, no measurable H_2 or O_2 species were detected over the single-component catalysts $\text{Co}_{\text{SAs}}@CNN$ or PtCo@CNN , demonstrating the dependence on the synergistic cooperation between single-atom Co and PtCo alloy composites supported on nanosheet CNN.

We observed that the H_2 and O_2 evolution rates over the $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ catalyst (see Figure S12) reached up to $85.7 \mu\text{mol/h}\cdot\text{g}$ and $21.3 \mu\text{mol/h}\cdot\text{g}$ under visible-light irradiation ($\lambda > 400 \text{ nm}$), showing the potential utilization of visible irradiation.³⁴ The apparent quantum efficiency (AQE) of product formation by $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ is relatively consistent with the optical absorption spectrum of $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$. Figure S13 shows the optical absorption spectrum of the $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ catalyst irradiated at different wavelengths, suggesting that the overall water splitting reaction is driven by absorption and photoexcitation. The performance of $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ is superior to those of most previously reported amorphous C_3N_4 -based materials (Table S4), especially the normalization activity of loaded metals ($48.30 \mu\text{mol/h}\cdot\text{mg}$), which is much larger than those of other reported amorphous C_3N_4 -based catalysts. We also carried out half cycle reactions for water under $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ in the presence of triethanolamine (TEOA) as the

sacrificial agent under visible-light irradiation ($>400 \text{ nm}$, see Figure 4d). This reaction exhibits a high H_2 production rate of $5.58 \text{ mmol/h}\cdot\text{g}$. The O_2 evolution reaction was tested with AgNO_3 (as the electron sacrificial reagent) and La_2O_3 (pH stabilizer). The $20 \text{ mg Co}_{\text{SAs}}/\text{PtCo@CNN}$ catalyst exhibited a moderate O_2 evolution rate of $3.8 \mu\text{mol/h}$, which was $1.4\times$ higher than that of $\text{Co}_{\text{SAs}}@CNN$ (Figure S14). Moreover, $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ greatly improved the degradation efficiency of an organic compound like rhodamine B (RhB) (enhanced about $24\times$, Table S1).

To better understand the relationship between the catalytic activity and the synergistic effect between Co SAs and the PtCo alloy of the $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ catalyst, we performed near-atmospheric X-ray photoelectron spectroscopy (NAP-XPS) and density function theory (DFT) calculations. After introducing $10 \text{ mbar H}_2\text{O}$ gas, the peak of Co^{2+} shifted positively by 0.5 eV , while no obvious shifts were observed for the peaks of Co^0 and Pt^0 (Figure S15a and b). The positive shift suggested that there was electron transfer to the adsorbed H_2O , indicating that the Co_{SAs} sites in $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ sample were responsible for H_2O activation. Meanwhile, $\bullet\text{OH}$ signal was only detected over the $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ systems and not over the $\text{Co}_{\text{SAs}}@CNN$ or PtCo@CNN catalysts (Figures 4a and S6). The combined NAP-XPS analysis and in situ EPR results suggested that Co_{SAs} sites acted as active sites for H_2O activation and that a spillover phenomenon of hydroxyl groups occurred between the Co SAs and the PtCo alloy. The relationship between the structures of the interfacial coordination sites and the energetics underlying reaction pathways is a determinant of the magnitude of this activity.^{5,35,36} Figure S16a and b supports the concept of favorable electronic orbital overlap between Co_{SAs} and PtCo that is coordinated by N atoms. We note that the concentration of negative charge on Co_{SAs} is conducive to carrying out the HER.³⁷ A reaction pathway of $\text{Co}_{\text{SAs}}/\text{PtCo@CNN}$ toward the HER is proposed in Figure 5a. The Gibbs free energy change of H adsorption (ΔG_{H^*}) on Co_{SAs} and the PtCo alloys supported on the CNN nanosheet was calculated to determine the configuration of the HER active site. It is well-known that the ΔG_{H^*} of an ideal HER catalyst should be close to 0 ($\Delta G_{\text{H}^*} \rightarrow 0$), which suggests that the catalysts are hard to poison.³⁸ For PtCo alloys, the high positive ΔG_{H^*} of the Co site and Pt sites (0.99 and 0.84 eV) imply that the HER on the alloy surface is nonspontaneous and needs to absorb a lot of energy to initiate. For Co SAs, the near-zero ΔG_{H^*} implies that the Co site in the form of a single atom can act as a highly active site for the HER that is comparable to or even better than Pt (111) (-0.03 versus -0.15 eV , respectively,

see Figure 5a). Therefore, the alloy surfaces are excluded from being candidates for HER active sites and the HER is evidently prone to occur on Co_{SAs} .

Next, we evaluated the possibility that the oxygen evolution reaction (OER) occurred on Co_{SAs} and also on PtCo alloys supported by the CNN nanosheet. As shown in Figure 5b, the suitable ΔG_{OH^*} values of the Pt and Co sites (0.04 and 0.07 eV, respectively) on the alloy surface imply that they both can be viewed as ideal candidates for OER active sites due to dual advantages of easy desorption and easy adsorption. In contrast, the ΔG_{OH^*} for the adsorption OH is negative for Co_{SAs} (−0.35 eV). The strong binding between HO and single-atom Co atoms allows the catalysts to be easily poisoned, which is antithetical to the production of a good OER activity. Subsequently, the second step ($\text{OH}^* \rightarrow \text{O}^*$) becomes the rate-limiting steps for the OER process. The ΔG_{O^*} values for this step are 2.18, 1.82, and 2.52 eV for the Pt site on the alloy surface, the Co site on the alloy surface, and the Co_{SAs} site, respectively. For the Co site on the PtCo alloy surface, each of the elementary steps is uphill, and its rate-limiting step is substantially available to complete the OER half-cell reaction.³⁹ As mentioned above, this analysis provides an explanation for our experimental observation of synergistic activity by the $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ catalyst. The synergy occurs through Co SAs-coordinated N atoms, which act as highly active sites for the HER half-reaction. Simultaneously, PtCo alloy nanoparticles, supported by nanosheet C_3N_4 , were configured to act as highly active sites for the OER.

3. CONCLUSION

In summary, we designed a novel two-component synergistic photocatalyst, $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$, that contained Co SAs and PtCo alloy nanoparticles supported over nanosheet $\text{g-C}_3\text{N}_4$. This catalyst efficiently facilitates the reaction for photocatalytic overall water splitting. Benefiting from a large surface area and a high pore volume, the nanosheet C_3N_4 possesses an excellent affinity for the activation of water molecules and provides abundant N coordination for the formation of Co SAs. The synergistic activity of Co SAs and PtCo alloys produces a bridge across the gap between maximum atom utilization and bifunctional reactivity for hydrogen and oxygen evolution. It also permits the transport of reactive intermediates over reaction centers and facilitates the separation of photogenerated electron–hole pairs. As a result, the $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ catalyst exhibits a high activity up to 300.9 $\mu\text{mol}/\text{h}\cdot\text{g}$ for H_2 production in the overall water splitting reaction and an activity of 5.58 $\text{mmol}/\text{h}\cdot\text{g}$ for H_2 production in the presence of triethanolamine (TEOA) under visible-light irradiation. This study not only presents a promising strategy for the development of efficient single-atom photocatalysts through the construction of synergistic alloy sites but also provides insights into structure–activity relationships for the overall water splitting reaction by photocatalytic processes.

4. EXPERIMENTAL SECTION

Synthesis. Bulk graphitic carbon nitride ($\text{g-C}_3\text{N}_4$) was prepared by the calcination of melamine (10 g) at 823 K (heating rate of 2.3 K/min) in a muffle furnace for 4 h. As shown in Figure 1a, nanosheet $\text{g-C}_3\text{N}_4$ was obtained following thermal exfoliation at 773 K (heating rate of 5 K/min) for 5 h in static air. The resulting $\text{g-C}_3\text{N}_4$ (1.2 g) had a high surface area and a large pore volume. It should be emphasized that the

specific pore volume is closely related to the selected crucible diameter. We chose a customized quartz crucible with a diameter of 10 cm.

To prepare the $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ sample, the carrier precursor (0.2 g of nanosheet $\text{g-C}_3\text{N}_4$) was first dispersed in H_2O (20 mL), and the solution was kept under high-power sonication for 6 h. Meanwhile, the mixture of 3 mg of Pt as H_2PtCl_6 and 1 mg of Co as $\text{Co}(\text{NO}_3)_2$ was prepared in an aqueous solution and added to the 10 mL of ethylene glycol (EG). The reaction mixture was then stirred for 2 h at 323 K under high vacuum (see Figure 1a). Then, the light-green EG solution containing the Pt and Co species was added into the carrier solution (1.5 wt % based on the Pt atomic weight and 0.5 wt % based on the Co atomic weight). The mixture was irradiated under UV–vis ($\lambda > 300$ nm, the information on 300 W Xe is shown in Figure S17) under high vacuum for 2 h to adequately reduce the noble metal cations with the help of triethanolamine (TEOA, 10% volume). Finally, the powder was collected by filtration, washed first with distilled water and then with ethanol, and finally dried at 333 K overnight. A Pt/PtCo@CN sample was produced under the same conditions to serve as a control. The difference between the $\text{Co}_{\text{SAs}}/\text{PtCo}@ \text{CNN}$ and Pt/PtCo@CN samples was that nanosheet $\text{g-C}_3\text{N}_4$ with a high surface area was replaced by bulk $\text{g-C}_3\text{N}_4$ in the latter.

The Pt/PtCo@CNN sample was processed through 3 wt % Pt and 1.5 wt % Co loading, which was accomplished by soaking the aqueous solutions of nanosheet $\text{g-C}_3\text{N}_4$ in H_2PtCl_6 and $\text{Co}(\text{NO}_3)_2$ under sonication for 30 min. The residual solvent was then evaporated in a water bath at 80 °C. The powder product was transferred to a covered crucible, heated to 180 °C at a rate of 5 °C/min in a muffle furnace, and held at that temperature for 1 h in air under atmospheric pressure. The furnace was allowed to cool naturally to room temperature before the Pt/PtCo@CNN powder sample was removed. The resulting sample was washed first with distilled water and then with ethanol, then dried at 60 °C overnight.

To synthesize the $\text{Co}_{\text{SAs}}@ \text{CNN}$ sample, precursors were first processed by high-power sonication. The cobalt precursors ($\text{Co}(\text{NO}_3)_2$ in aqueous solutions) were stirred with CNN overnight, then washed with ultrapure water. Products were subsequently obtained by photoreduction under vacuum with the help of triethanolamine (TEOA, 10% volume).⁴⁰ For the synthesis of the PtCo@CNN sample, the PtCo alloy was first synthesized via the combined reduction of Pt and Co salts in ethylene glycol (EG, pH 11.5 by the addition of NaOH solution). The mixture was stirred for 3 h to form a homogeneous and light green solution to construct the PtCo alloys. Then, to the alloy solution was added 0.2 g of CNN powder, followed by ultrasonic dispersion. The sample was then transferred to a Teflon-lined stainless-steel autoclave and heated at 200 °C for 6 h.²⁸

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.2c00751>.

Materials characterizations, ^1H NMR spectra, photocatalytic H_2 and O_2 evolution, and theoretical configurations (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the National Natural Science Foundation of China (Grants 22072045, 21603073, and 21574043), the Ministry of Science and Technology of the People's Republic of China (Grant 2018YFC1602800), Xing-Fu-Zhi-Hua Foundation of ECNU, and Microscale Magnetic Resonance Platform of ECNU.

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